

03-10-2025

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \underline{V}_{\beta})$$

↳ look to notes for est.

$$H_0: \beta_1 = 0$$

$$t = \frac{\hat{\beta}_1}{\text{s.e.}(\hat{\beta}_1)} = \frac{\sqrt{n}(\hat{\beta}_1 - 0)}{\sqrt{\hat{V}_{\beta_{1,1}}}}$$

↳ look to notes for behavior

Significance level: pick willingness to make error (.05)

P-value:

→ Pr (getting a stat/est of β as large in abs. value as we did if H_0 were true)

$$\hookrightarrow p = \Pr(T > |t| \mid H_0 \text{ is true})$$

$\uparrow$
 $t \in T$

$$\hookrightarrow T \sim N(0, 1)$$

$$= \Pr(\overset{\uparrow}{z} > |t|) = \Pr(z < -|t|) + \Pr(z > |t|)$$

$$= 2\Pr(z < -|t|) = 2\Phi(-t)$$

$\downarrow$

in R: `pnorm(-abs(t)) * 2`

CDF of Standard normal

Confidence Interval:

$$\hookrightarrow \hat{C} = [\hat{\theta} \pm c_{1-\alpha/2} \text{s.e.}(\hat{\theta})]$$

$\hookrightarrow 95\%$ chance range includes $\hat{\theta}$

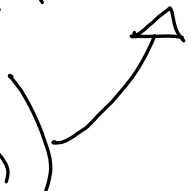
Wald - Stat:

$\hookrightarrow$ might want test for multiple hypotheses

$$\text{i.e.: } H_0: \beta_1 = \beta_2 = 0 \quad // \quad H_0: \theta = \theta_0$$

$$\begin{aligned} \hookrightarrow W &= n(\hat{\theta} - \theta_0)^T \hat{V}_0^{-1} (\hat{\theta} - \theta_0) \\ &= \left(\sqrt{n} (\hat{\theta} - \theta_0)^T \hat{V}_0^{-\frac{1}{2}} \right) \left(\hat{V}_0^{-\frac{1}{2}} \sqrt{n} (\hat{\theta} - \theta_0) \right) \end{aligned}$$

if H_0 true
as $n \rightarrow \infty$
 $Z \sim N(0, I)$



$\hookrightarrow$ if H_0 : true

$$W \rightarrow Z^T Z = \sum_{i=1}^q Z_i^2 \sim \chi_q^2$$

$\hookrightarrow$ if H_0 : false

$$W \rightarrow \infty$$

$\hookrightarrow$ turn into P:

$$P = 1 - G_q(W) \quad \text{s.t. } G_q \text{ is CDF of } \chi^2 \text{ random variable}$$

$$Ex: W = 7, q = 2$$

R:

$$p \leftarrow 1 - \text{pchisq}(7, df = 2)$$

$$p = 0.03$$

↳ look up $G_q(W)$ CDF stuff

Callaway Questions:

↳ stuff to improve on stat / Lin AI?

↳ study strategy?

↳ biggest place to improve?

Functions of Parameters:

$$\hat{\theta} = r(\hat{\beta}) ; r: \mathbb{R}^k \rightarrow \mathbb{R}^q$$

$$\theta = r(\beta)$$

Ex:

$$r(\beta) = \beta_1$$

$$r(\beta) = \frac{\beta_j}{\beta_2}$$

↳ asymptotic dist. of $\hat{\theta}$

→ consistency: $\hat{\theta} \xrightarrow{p} \theta$ if r is cts at β by CMT

↳ ① est. $\hat{\beta}$

↳ ② calc $\hat{\theta}$

→ asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma_\theta)$ by delta method.

Delta Method:

Case 1: r is a linear function:

$$\theta = r(\beta) = \underline{R}^T \beta \quad \text{s.t. } \underline{R}_{k \times q}$$

$$\hat{\theta} = \underline{R}^T \hat{\beta}$$

$$\hookrightarrow \text{ex: } \underline{R} = \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} \Rightarrow \text{choose } R,$$

$$\sqrt{n}(\underline{R}^T \hat{\beta} - \underline{R}^T \beta) = \underline{R}^T \underbrace{\sqrt{n}(\hat{\beta} - \beta)}_{\xrightarrow{d} N(0, \underline{V}_\beta)} \xrightarrow{d} N(0, \underline{V}_\theta)$$

$$\text{s.t. } \underline{V}_\theta = \underline{R}^T \underline{V}_\beta \underline{R}; \quad \underline{V}_\theta q \times q$$

Case 2: Next class.

$$\text{ex: } \underline{R} = \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix}; \quad \underline{V}_\theta = \underline{R}^T \underline{V}_\beta \underline{R} = (v_{11} \dots v_{1k}) \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} = v_{11}$$

$$\underline{V}_\beta = \begin{bmatrix} v_{11} & \dots & v_{1k} \\ \vdots & \ddots & \vdots \\ v_{k1} & \dots & v_{kk} \end{bmatrix}$$